Fair Comparison of Russia–Ukraine Equipment Loss Rates: From Estimation Bias to Interval Estimation

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Abstract

Official battlefield reports from Russia and Ukraine differ sharply, making direct comparisons of equipment losses vulnerable to reporting incentives and observation bias. This paper uses bilateral Oryx visually verified equipment-loss records, covering 23,556 Russian losses and 11,397 Ukrainian losses from February 2022 to June 2026, together with WarSpotting item-level records and Ukrainian General Staff claims, to construct a statistical framework for point estimation, interval estimation, significance testing, and observation-bias sensitivity analysis. Under a Poisson mean-rate model, the estimated verified daily loss rates are 15.11 items per day for Russia and 7.31 items per day for Ukraine, yielding an observed rate ratio of 2.07 with a 95% confidence interval of [2.02, 2.11]. An exact conditional test for two independent Poisson means rejects equality at $p < 0.0001$. Because daily loss counts are temporally dependent, the standard error from Moving Block Bootstrap is about 2.4 times the IID Bootstrap estimate, showing that independence-based uncertainty estimates are too narrow. Finally, we incorporate asymmetric visibility probabilities into a sensitivity analysis. Under the specified prior assumptions for observation probabilities, the bias-corrected rate ratio has a median of 1.63 and a 95% interval of [0.67, 3.44], with $P(\rho_{\text{corrected}} > 1) = 88.1\%$. The evidence supports the directional conclusion that Russia's visually verified loss rate is higher, but the strength of this conclusion depends on assumptions about differential observability. A fair comparison therefore requires not only point estimates and significance tests, but also interval estimates and explicit bias sensitivity analysis.

Keywords: Poisson distribution; maximum likelihood estimation; confidence interval; block bootstrap; two-sample comparison; exact Poisson test; conflict data

1 Introduction

1.1 Background

Since Russia's full-scale invasion of Ukraine on February 24, 2022, both sides have published regular battlefield reports. These official claims differ substantially from one another and from third-party verification. For example, Ukrainian official claims for Russian tank losses are far larger than the number of tanks independently verified by images. This discrepancy raises a statistical problem: how can one estimate and compare true equipment loss rates when all available data sources are affected by reporting bias, selection bias, and heterogeneous observation probabilities?

Open-source intelligence (OSINT) platforms such as Oryx provide a useful benchmark. Oryx records visually verified losses using public photographs and videos, and it documents losses for both Russia and

Ukraine. These data are not the same as true losses: a loss must be photographed, uploaded, discovered, classified, and verified before it appears in the dataset. The probability of observation varies across territory, equipment type, loss status, time period, and side. Nevertheless, visually verified data are more auditable than official claims and provide a defensible basis for statistical inference.

1.2 Research Questions

This paper addresses three connected questions:

1. How can equipment loss rates be estimated with point estimates and confidence intervals?
2. Are the verified loss rates of the two sides statistically different?
3. How sensitive are the conclusions to asymmetric observation probabilities?

1.3 Contributions

This work applies a coherent statistical chain to bilateral equipment losses: Poisson maximum likelihood estimation (MLE), Wald/Score/Exact confidence intervals, Moving Block Bootstrap and BCa intervals for temporal dependence, exact conditional tests for two independent Poisson rates, stratified comparisons by equipment type and phase, and a bootstrap-based sensitivity analysis for observation-probability bias. The main contribution is not only the numerical comparison of the two sides, but also a reproducible framework for moving from observed OSINT records to uncertainty-aware and bias-aware conflict-data inference.

2 Data

2.1 Data Sources

Three complementary data sources are used.

Source	Type	Content	Period	Size
Oryx bilateral data (leedrake5/Russia-Ukraine)	Visually verified, bilateral	Daily cumulative Russian and Ukrainian losses, including type and status breakdowns	2022-02-25 to 2026-06-04	1,559 days
WarSpotting (Kaggle: zsoltlazar)	Visually verified, item-level	Russian equipment-loss records with coordinates and model information	2022-02-24 to 2026-06-03	22,970 items
Ukrainian General Staff claims (Kaggle: piterfm)	Official claims	Daily cumulative Russian losses by category	2022-02-25 to 2026-05-31	1,557 days

The bilateral Oryx data are taken from the public leedrake5 Google Sheet scraper. WarSpotting and Ukrainian General Staff data are obtained from Kaggle or kagglehub sources. Oryx totals are treated as visually verified losses, not as true total losses.

2.2 Bilateral Oryx Overview

As of June 4, 2026, Oryx verifies 23,556 Russian equipment losses and 11,397 Ukrainian equipment losses, an observed ratio of 2.07:1 over 1,559 days.

Status	Russia	Ukraine	Russia share	Ukraine share
Destroyed	~19,625	~9,627	78.8%	77.8%
Captured	~3,520	~1,540	12.0%	10.4%
Abandoned	~1,542	~797	5.1%	5.9%
Damaged	~1,033	~715	4.2%	5.9%

Russian captured-equipment counts are larger both absolutely and as a share of total losses, consistent with major Russian withdrawals and equipment abandonment in 2022.

Equipment type	Russia	Ukraine	Ratio (RU/UA)
Armoured fighting vehicles (AFV)	2,682	597	4.49
Infantry fighting vehicles (IFV)	6,702	1,641	4.08
Vehicles	4,949	1,456	3.40
Tanks	4,729	1,504	3.14
Logistics vehicles	1,094	391	2.80
Engineering vehicles	759	299	2.54
Aircraft	529	281	1.88
Air-defense systems	603	394	1.53
Artillery	1,762	1,202	1.47
Armoured personnel carriers (APC)	**804**	**1,425**	**0.56**

The type-level counts use a daily-difference exploratory convention for subcategory cumulative columns. Because Oryx sometimes retroactively revises classifications, these subcategory ratios should be interpreted as structural evidence rather than exact item-count totals.

2.3 Observation Bias

Oryx records a visually verified subset of losses. It is affected by several observation mechanisms:

1. **Spatial bias.** Losses in Ukrainian-controlled areas, near roads, or near populated locations are more likely to be photographed and uploaded. Russian losses in Ukrainian-controlled territory may therefore be observed more easily than Ukrainian losses in Russian-held territory.
2. **Equipment bias.** Large platforms such as tanks and artillery are easier to identify than small drones or portable systems.
3. **Status bias.** Destroyed equipment is easier to classify than lightly damaged equipment.
4. **Temporal bias.** Social-media activity, front-line stability, and volunteer attention vary over time.

These mechanisms mean that the observed ratio may be biased relative to the true loss-rate ratio. A fair comparison must therefore report both statistical uncertainty and sensitivity to observation assumptions.

3 Methods

3.1 Poisson Model and MLE

Let Y_t be the number of losses on day t. As a first-order model, assume

$$Y_t \sim \text{Poisson}(\lambda),$$

where λ is the mean daily loss rate. For independent observations $Y_1, \ldots, Y_n$, the likelihood is

$$L(\lambda) = \prod_{t=1}^{n} \frac{e^{-\lambda}\lambda^{Y_t}}{Y_t!}.$$

The log-likelihood is

$$\ell(\lambda) = -n\lambda + \left(\sum_{t=1}^{n} Y_t\right) \log \lambda - \sum_{t=1}^{n} \log(Y_t!),$$

and the MLE is

$$\hat{\lambda}_{\text{MLE}} = \frac{1}{n}\sum_{t=1}^{n} Y_t = \bar{Y}.$$

For the Russia-to-Ukraine rate ratio $\rho = \lambda_{RU}/\lambda_{UA}$, the MLE is $\hat{\rho} = \hat{\lambda}_{RU}/\hat{\lambda}_{UA}$. A delta-method standard error is

$$SE(\hat{\rho}) = \hat{\rho}\sqrt{\frac{1}{\hat{\lambda}_{RU} n} + \frac{1}{\hat{\lambda}_{UA} n}}.$$

3.2 Poisson Confidence Intervals

This paper compares three Poisson confidence-interval methods.

Wald interval. The asymptotic interval is

$$\hat{\lambda} \pm z_{\alpha/2}\sqrt{\frac{\hat{\lambda}}{n}}.$$

It is simple but can under-cover when λ is small.

Score interval. Solving the score equation yields

$$\lambda = \hat{\lambda} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{\lambda}}{n} + \frac{z^2}{4n^2}}.$$

Exact (Garwood) interval. Using the relationship between Poisson counts and the chi-square distribution,

$$\lambda_L = \frac{1}{2n}\chi^2_{2Y,\alpha/2}, \qquad \lambda_U = \frac{1}{2n}\chi^2_{2(Y+1),1-\alpha/2}.$$

Exact intervals are conservative but maintain nominal coverage for rare-event settings.

3.3 Moving Block Bootstrap and BCa

Daily equipment losses exhibit temporal dependence. The estimated lag-1 autocorrelation for Russian losses is 0.589. IID bootstrap resampling therefore understates uncertainty. Moving Block Bootstrap (MBB) preserves local dependence by resampling overlapping time blocks. The block length is selected from the empirical autocorrelation decay rule, giving $b = 11$; the heuristic $n^{1/3}$ gives a similar value.

BCa intervals further adjust percentile intervals for bias and acceleration:

$$\alpha_1 = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_{\alpha/2}}{1 - \hat{a}(\hat{z}_0 + z_{\alpha/2})}\right), \quad \alpha_2 = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_{1-\alpha/2}}{1 - \hat{a}(\hat{z}_0 + z_{1-\alpha/2})}\right).$$

3.4 Exact Test for Two Independent Poisson Rates

For independent samples

$$Y_{1i} \sim \text{Poisson}(\lambda_1), \quad Y_{2j} \sim \text{Poisson}(\lambda_2),$$

we test

$$H_0 : \lambda_1 = \lambda_2 \quad \text{against} \quad H_1 : \lambda_1 \neq \lambda_2.$$

Conditional on the total count S, the group-1 count follows

$$Y_1 = \sum_i Y_{1i} \mid S \sim \text{Binomial}\left(S, \frac{n_1}{n_1 + n_2}\right).$$

The resulting two-sided exact p-value provides a finite-sample comparison of two Poisson rates.

3.5 Observation-Probability Bias Correction

Let observed verified losses and true losses satisfy

$$Y_{\text{obs}} = pY_{\text{true}},$$

where $p \in (0, 1]$ is the probability that a true loss is photographed, uploaded, discovered, and verified. For the rate ratio,

$$\rho_{\text{true}} = \frac{\lambda_{RU}^{\text{true}}}{\lambda_{UA}^{\text{true}}} = \frac{\lambda_{RU}^{\text{obs}}/p_{RU}}{\lambda_{UA}^{\text{obs}}/p_{UA}} = \rho_{\text{obs}}\frac{p_{UA}}{p_{RU}}.$$

Thus, the observed rate ratio is unbiased for the true rate ratio only if $p_{RU} = p_{UA}$. If Ukrainian losses are harder to observe, $p_{UA} < p_{RU}$ and the true ratio is lower than the observed ratio.

Because p_{RU} and p_{UA} are not identified from Oryx alone, we report sensitivity intervals:

$$\rho_{\text{true}} \in \left[\rho_{\text{obs}}\frac{p_{UA}^{\text{min}}}{p_{RU}^{\text{max}}}, \rho_{\text{obs}}\frac{p_{UA}^{\text{max}}}{p_{RU}^{\text{min}}}\right].$$

We also embed observation probabilities into a bootstrap procedure using

$$p_{RU} \sim \text{Beta}(7, 3), \qquad p_{UA} \sim \text{Beta}(5.5, 4.5).$$

These priors encode the assumption that Russian losses are, on average, more observable than Ukrainian losses because many Russian losses occur in Ukrainian-controlled territory.

4 Empirical Results

4.1 Overall Rate Estimates

Parameter	Russia	Ukraine
Mean daily verified loss rate $\hat{\lambda}$	15.110	7.310
Standard error	0.098	0.068
95% exact CI	[14.917, 15.304]	[7.177, 7.446]
Cumulative verified losses	23,556	11,397

The observed rate ratio is $\hat{\rho} = 2.067$ with standard error 0.024 and a 95% Wald interval of [2.02, 2.11]. The exact conditional test rejects equal rates with $p < 0.0001$. Figure 1 shows that Russia's verified losses were much higher in the early war but the gap narrowed over time. Figure 2 shows clearly separated rate intervals.

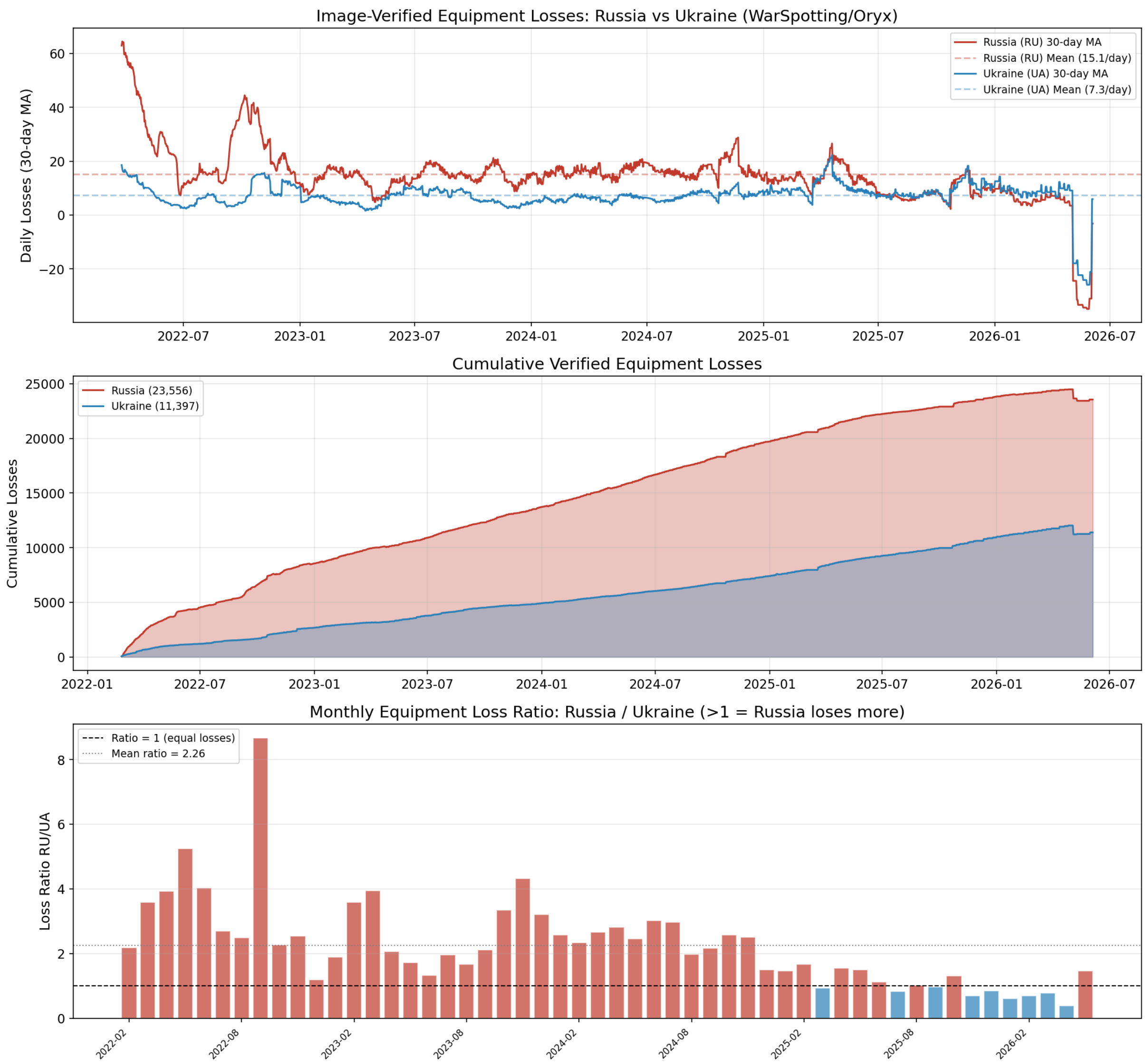

Figure 1: Daily and monthly Oryx loss trends for both sides.

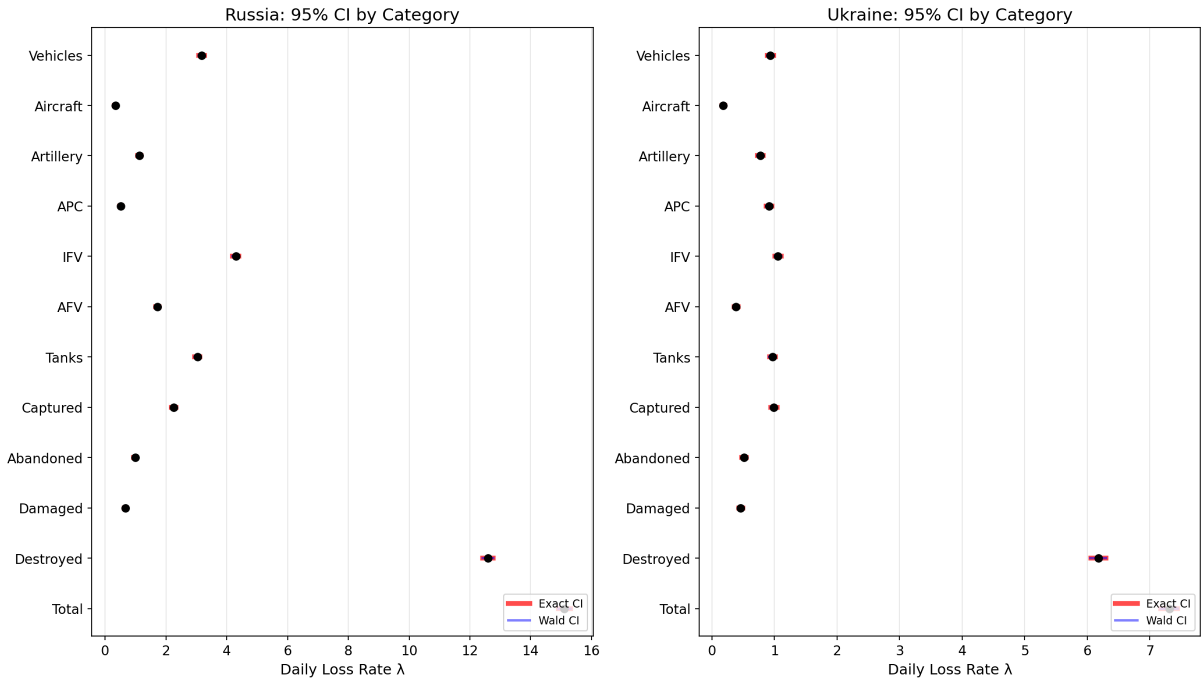

Figure 2: Confidence intervals for verified daily loss rates of both sides.

4.2 Monthly Dynamics

Monthly aggregation reveals strong time variation. In the early phase from February to August 2022, the Russia-to-Ukraine ratio was above 3. During the Ukrainian counteroffensives of late 2022, the ratio declined toward 1.5–2.0. During 2023–2024, the ratio fluctuated around 1–2. In 2025–2026, several windows approached or fell below 1. If months affected by negative retroactive corrections are excluded, recent ratios are about 0.65–0.8; if all corrections are included, the weighted 2025–2026 ratio is about 0.96. The robust qualitative conclusion is that the verified loss-ratio advantage observed early in the war has narrowed sharply.

4.3 Equipment-Type Comparisons

Type	$\hat{\lambda}_{RU}$	$\hat{\lambda}_{UA}$	Ratio	E-test p	Bonferroni
AFV	1.720	0.383	4.49	< 0.0001	***
IFV	4.298	1.052	4.08	< 0.0001	***
Vehicles	3.174	0.934	3.40	< 0.0001	***
Tanks	3.033	0.965	3.14	< 0.0001	***
Logistics	0.702	0.251	2.80	< 0.0001	***
Engineering	0.487	0.192	2.54	< 0.0001	***
Aircraft	0.339	0.180	1.88	< 0.0001	***
Air defense	0.387	0.253	1.53	< 0.0001	***
Artillery	1.130	0.771	1.47	< 0.0001	***
APC	**0.516**	**0.914**	**0.56**	< 0.0001	***

All ten equipment-type comparisons remain significant after Bonferroni correction. Russia has especially high verified losses in offensive and maneuver equipment such as AFVs, IFVs, vehicles, and tanks. APC is the main reverse category, with higher verified Ukrainian losses. Figures 3 and 4 visualize these patterns.

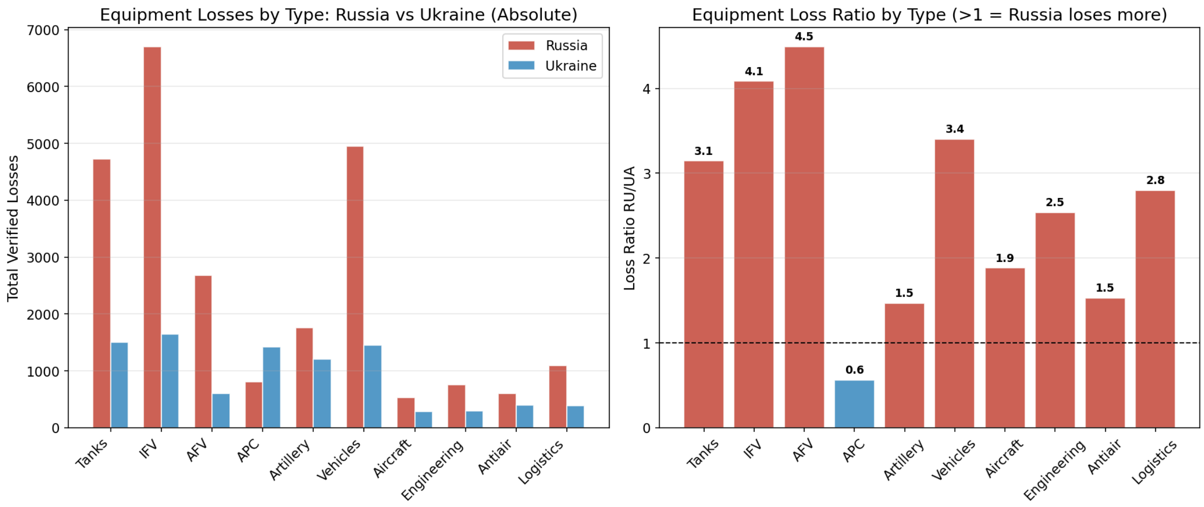

Figure 3: Equipment-type loss-rate comparison between Russia and Ukraine.

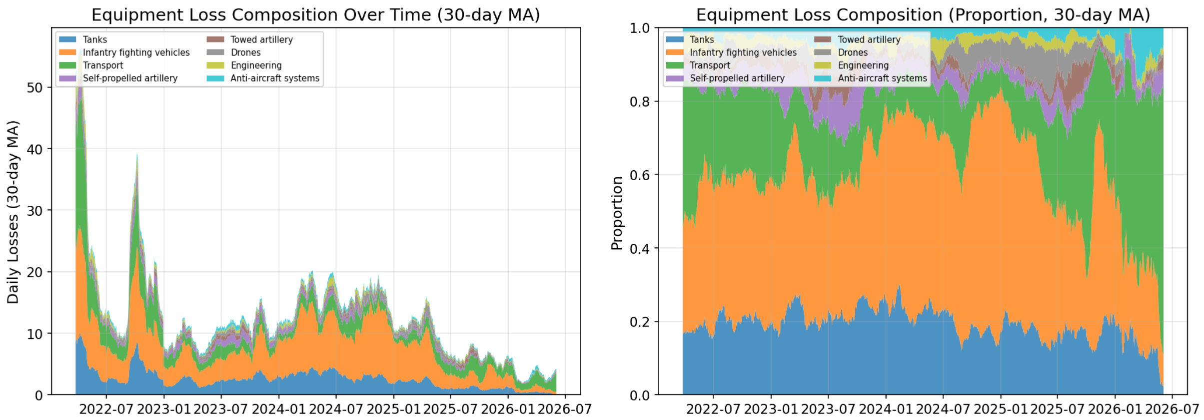

Figure 4: Russian equipment-type composition in WarSpotting records.

4.4 Loss Status

Status	$\hat{\lambda}_{RU}$	$\hat{\lambda}_{UA}$	Ratio	E-test p
Destroyed	12.59	6.17	2.04	< 0.0001
Captured	2.26	0.99	2.29	< 0.0001
Abandoned	0.99	0.51	1.93	< 0.0001
Damaged	0.66	0.46	1.44	< 0.0001

Destroyed equipment dominates both sides' verified records. Captured equipment is more prominent for Russia, consistent with episodes of withdrawal and abandonment. Figures 5 and 6 summarize these distributions.

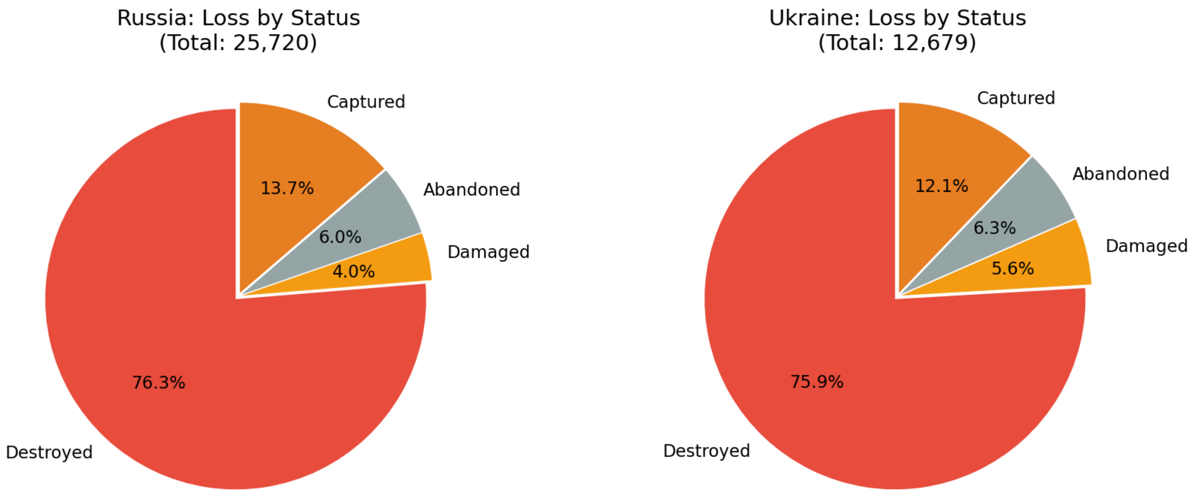

Figure 5: Loss-status composition for both sides.

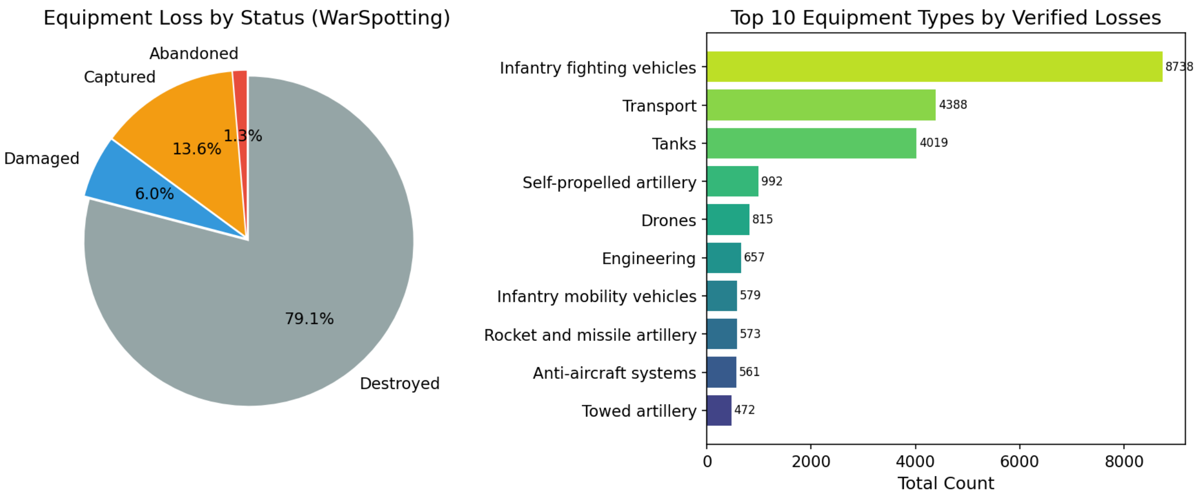

Figure 6: Russian loss status and equipment-type distribution in WarSpotting.

4.5 Confidence-Interval Coverage

Monte Carlo simulations with $N = 10{,}000$ replications and $n = 30$ days evaluate coverage for $\lambda \in \{0.5, 1, 2, 5, 10, 15, 20, 50\}$.

True λ	Wald	Score	Exact
0.5	**0.919** (low)	0.951	0.966
1.0	0.929 (low)	0.946	0.957
2.0	0.948 (slightly low)	0.954	0.961
5.0	0.955	0.956	0.956
10.0	0.950	0.947	0.950
15.0	0.949	0.949	0.953

Wald intervals under-cover for small rates, while Exact intervals maintain conservative coverage. For rare equipment categories, Exact intervals are therefore preferred. Figures 7–9 show interval estimates, interval-width comparisons, and Poisson Q-Q diagnostics.

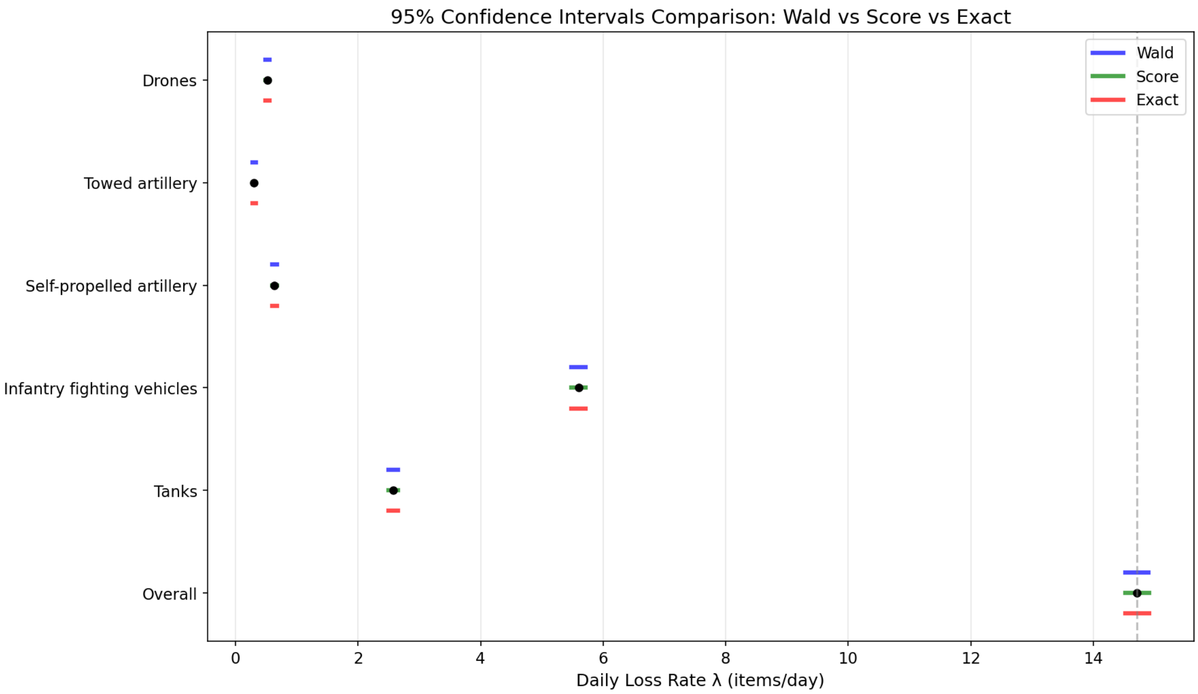

Figure 7: Poisson confidence-interval forest plot.

4.6 Block Bootstrap

Method	Bootstrap SE	95% interval
IID Bootstrap ($b = 1$)	0.42	[13.91, 15.54]
Block Bootstrap ($b = 11$)	**0.99**	[12.82, 16.59]
BCa Block Bootstrap ($b = 11$)	—	[13.18, 17.29]

The block-bootstrap standard error is about 2.4 times the IID bootstrap value. Ignoring temporal dependence therefore understates uncertainty by more than half. Figure 10 shows the broader block-bootstrap

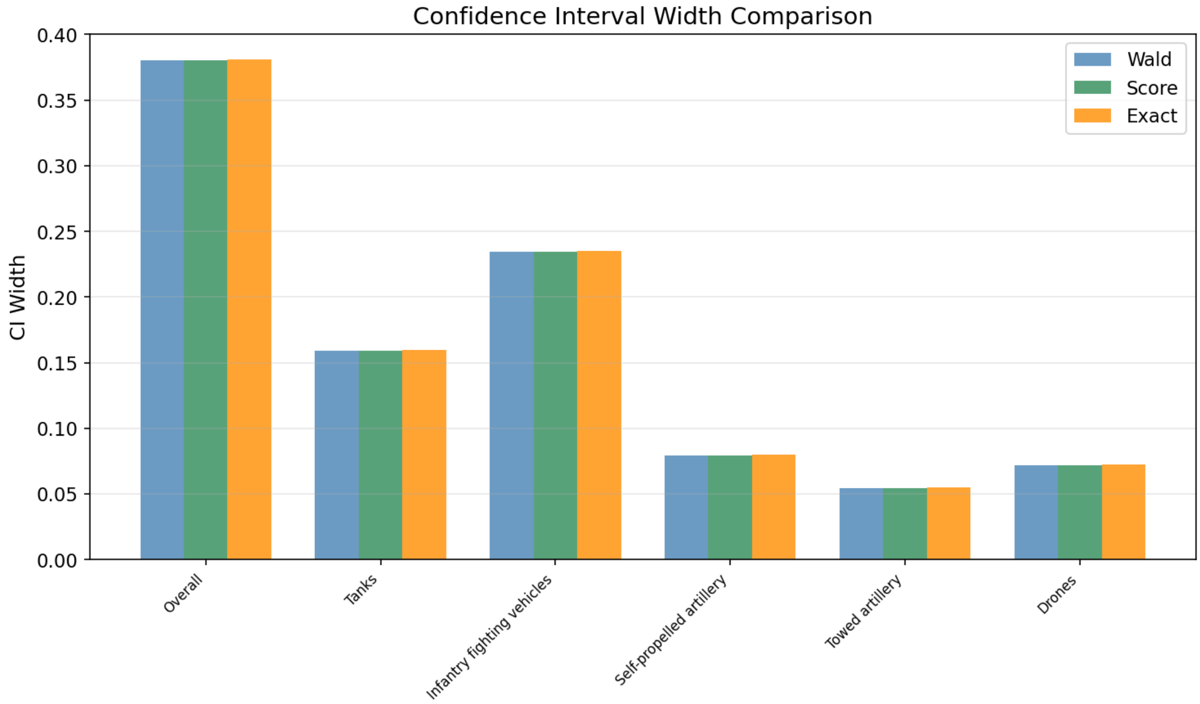

Figure 8: Interval-width comparison across confidence-interval methods.

distribution.

4.7 War Phases

Phase	Daily $\hat{\lambda}$	SE	Days
Phase 1 (2022.2–4), initial invasion	53.38	0.90	66
Phase 3 (2022.9–12), Ukrainian counteroffensive	24.75	0.45	122
Phase 7 (2024.3–7), spring offensive	17.65	0.34	153
Phase 9 (2025.1–2026.6), recent period	7.79	0.12	519

A likelihood-ratio test strongly rejects equal phase means ($LR = 8919.0$, $df = 8$, $p < 0.0001$). The early-war verified loss level is much higher than the recent level. Figures 11 and 12 show the phase structure.

4.8 Official Claims versus Verified Losses

Equipment	Verified rate	Claimed rate	Claim/verified ratio	E-test p
Tanks	2.57	7.68	**2.98**	< 0.0001
Armoured vehicles	5.60	15.85	**2.83**	< 0.0001
Artillery	0.94	27.62	**29.44**	< 0.0001

Official claims substantially exceed visually verified counts. The artillery ratio is especially large, likely reflecting both verification difficulty and differences in definitions such as suppressed or damaged positions versus destroyed systems. Figure 13 shows the divergence between official and verified series.

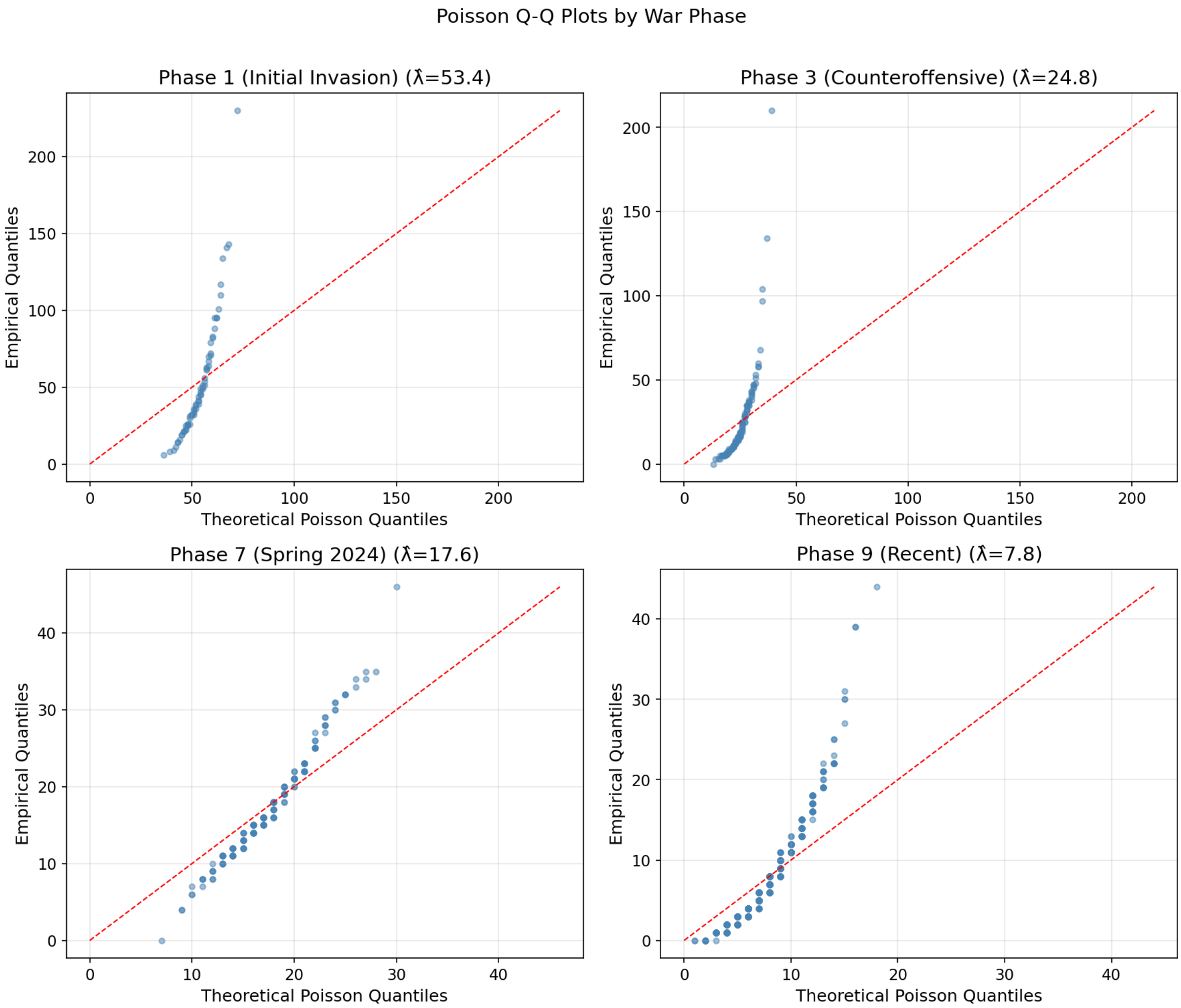

Figure 9: Poisson Q-Q diagnostic plot.

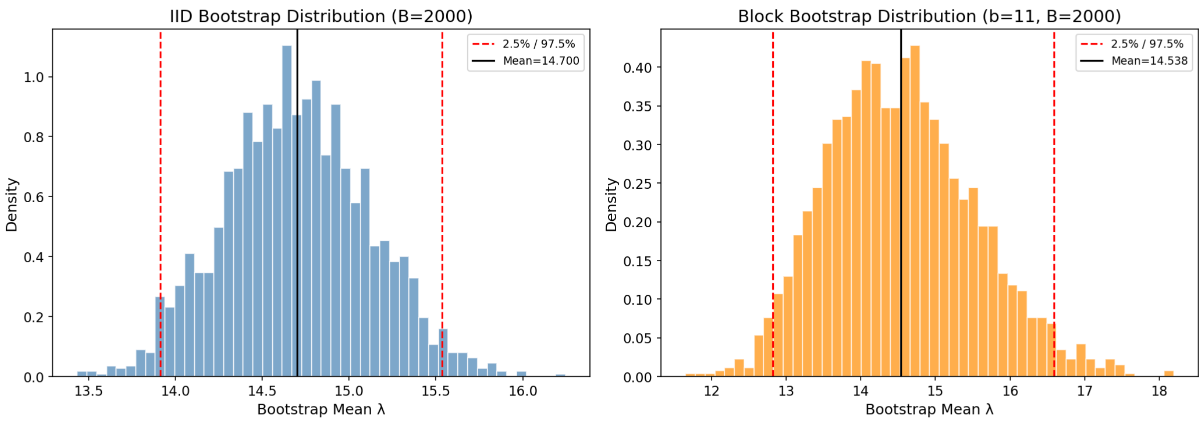

Figure 10: IID Bootstrap and Moving Block Bootstrap distributions.

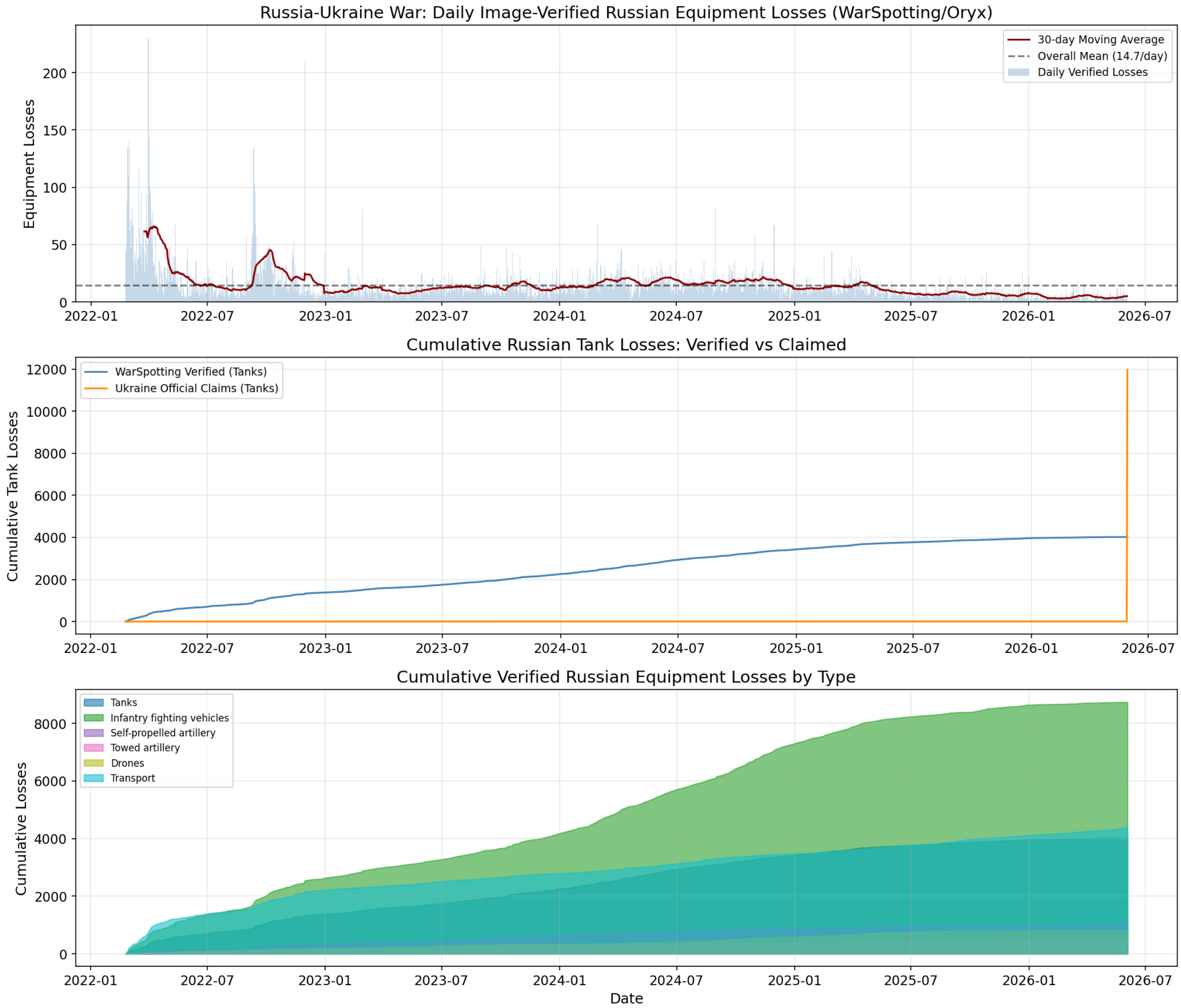

Figure 11: Russian daily verified loss time series from WarSpotting.

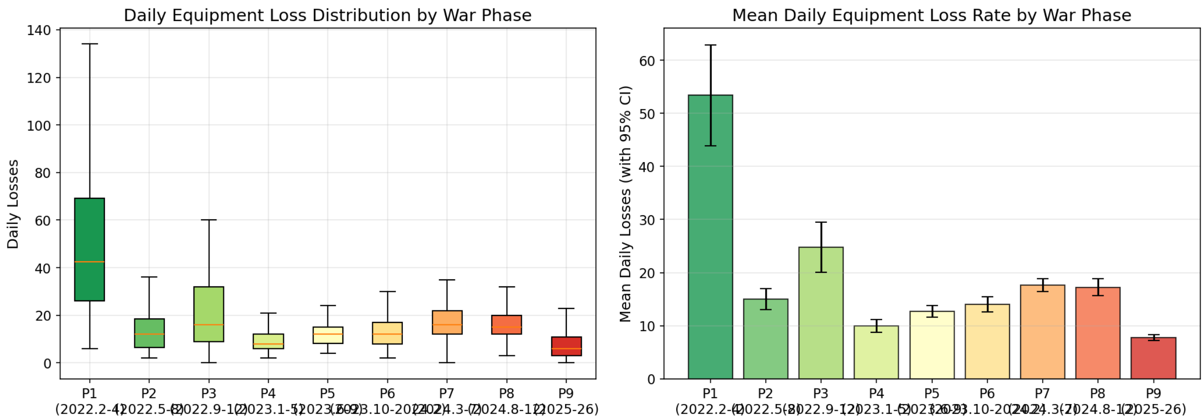

Figure 12: Daily loss distributions by war phase.

Figure 13: Ukrainian official claims versus WarSpotting verified records.

4.9 Bias-Corrected Rate-Ratio Inference

The observed rate ratio is about 2.06–2.07. A reversal of the conclusion $\rho_{\text{true}} > 1$ would require

$$\frac{p_{UA}}{p_{RU}} < \frac{1}{2.06} \approx 0.485.$$

Thus, Ukrainian losses would need to be observed at less than half the Russian observation probability for the rate-ratio conclusion to reverse.

Scenario	p_{RU} range	p_{UA} range	ρ_{true} range	Still > 1?
Conservative, large visibility gap	[0.8, 0.9]	[0.4, 0.5]	[0.92, 1.29]	Borderline
Moderate, plausible assumption	[0.7, 0.8]	[0.5, 0.7]	[1.29, 2.06]	Yes
Optimistic, small visibility gap	[0.5, 0.7]	[0.6, 0.8]	[1.77, 3.30]	Yes

Embedding $p_{RU} \sim \text{Beta}(7, 3)$ and $p_{UA} \sim \text{Beta}(5.5, 4.5)$ into the block-bootstrap procedure yields a bias-corrected median rate ratio of 1.63 and a 95% interval of [0.67, 3.44]. Under these priors, $P(\rho_{\text{corrected}} > 1) = 88.1\%$. The conclusion is therefore directionally robust but not entirely identified from Oryx data alone. Figure 14 displays the sensitivity surface.

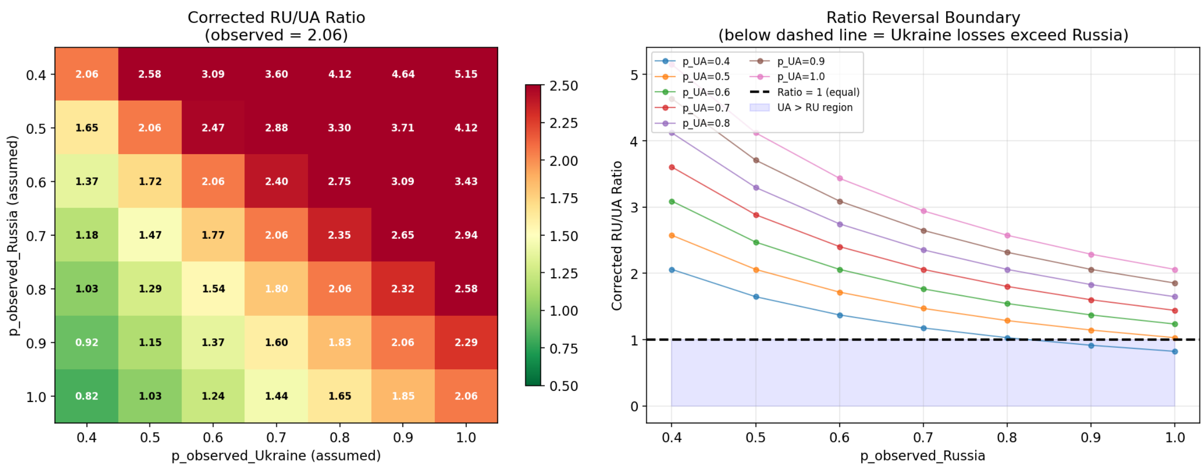

Figure 14: Sensitivity of the corrected rate ratio to observation probabilities.

4.10 Spatial Distribution

WarSpotting contains coordinates for 14,066 Russian loss records, about 61.2% of the item-level dataset. The eastern theater, especially Donetsk, dominates the spatial distribution. The hottest one-degree grid cell, approximately $(48°N, 37°E)$ around the Pokrovsk direction, contains 3,019 losses, or 21.5% of geocoded losses. The top three Donetsk-region grid cells together account for about 45.1%. Figure 15 shows the spatial concentration.

4.11 Weekly Bootstrap Check

Weekly aggregation provides an additional robustness check.

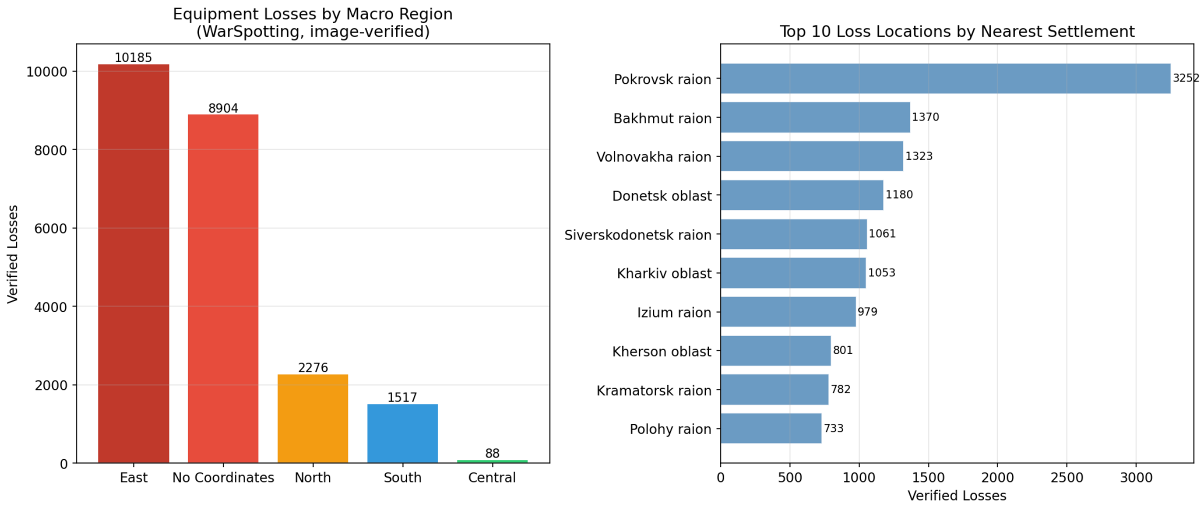

Figure 15: Spatial distribution of Russian equipment losses in WarSpotting.

Metric	Daily block bootstrap	Weekly bootstrap
Bootstrap SE	0.9868	**0.8212**
Percentile 95% CI	[12.82, 16.59]	[13.15, 16.37]
BCa 95% CI	[13.18, 17.29]	[13.31, 16.61]
BCa z_0	0.2211	0.0386
BCa a	0.0222	0.0406

The weekly bootstrap standard error is still much larger than the IID bootstrap estimate. This confirms that the uncertainty inflation caused by temporal dependence is not an artifact of daily frequency. Figure 16 shows the weekly-bootstrap distribution.

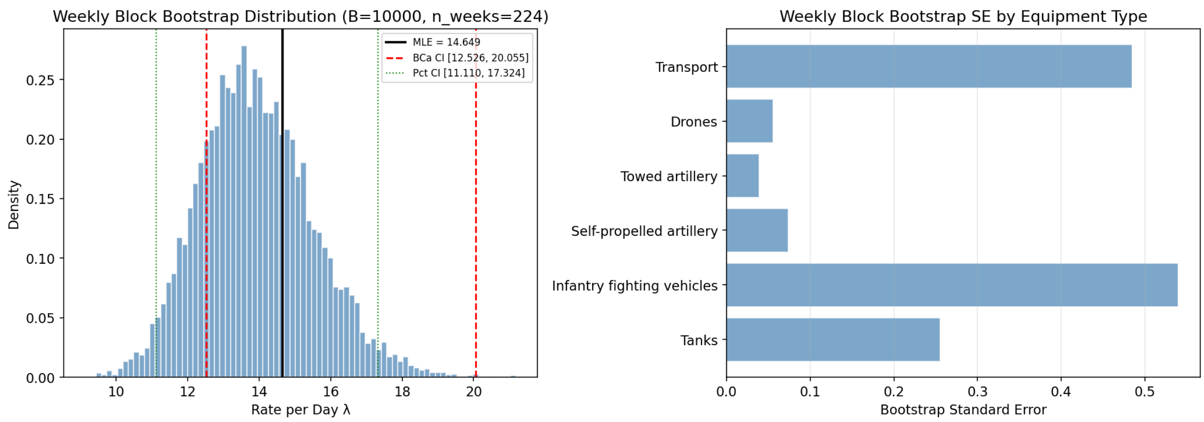

Figure 16: Weekly block-bootstrap distribution and BCa interval.

5 Discussion

5.1 Methodological Implications

First, exact Poisson intervals are preferable for rare equipment categories because Wald intervals undercover at small rates.

Second, temporal dependence is substantial in conflict data, so IID bootstrap or simple asymptotic standard errors can be misleadingly small.

Third, exact conditional tests provide a robust method for comparing two Poisson rates without relying entirely on large-sample approximations.

Fourth, observation-probability bias is not a nuisance detail; it is central to any fair comparison of visually verified conflict losses.

5.2 Limitations

The main limitation is that observation probabilities are not identified from Oryx alone. The bias-corrected results depend on prior assumptions about p_{RU} and p_{UA}.

The Poisson model is also a first-order approximation; real conflict data may be overdispersed, phase-dependent, and spatially clustered. Finally, Oryx bilateral data and WarSpotting item-level records use related but not perfectly identical classification conventions.

5.3 Future Work

Future work could use Bayesian hierarchical models to integrate multiple sources, including Oryx, official claims, satellite imagery, and other OSINT sources.

Negative-binomial or generalized Poisson models could handle overdispersion. Spatiotemporal point-process models could explicitly model front-line movement, battle intensity, and local autocorrelation.

The estimates in this paper can also serve as inputs for forecasting models of equipment attrition.

6 Conclusion

Using bilateral Oryx visually verified equipment-loss records, this paper estimates daily loss rates of 15.11 items per day for Russia and 7.31 items per day for Ukraine. The observed rate ratio is 2.07 with a narrow 95% confidence interval of [2.02, 2.11], and an exact two-Poisson conditional test rejects equal rates at $p < 0.0001$.

However, the observed Oryx ratio is not automatically a true loss ratio because observation probabilities differ across sides, territory, equipment types, and time. Under a prior-based observation-bias sensitivity analysis, the corrected ratio has a median of 1.63 and a wide 95% interval of [0.67, 3.44], with 88.1% of bootstrap-prior draws remaining above 1.

The main conclusion is therefore directional and conditional: visually verified data strongly indicate higher Russian verified loss rates, but fair inference requires explicit reporting of uncertainty and observation-bias sensitivity.

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